1D SDC as a Layered DAG

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October 2026

MFE and Thermodynamic Favourability

Minimum Free Energy of a 1D SDC - Dynamic Programming

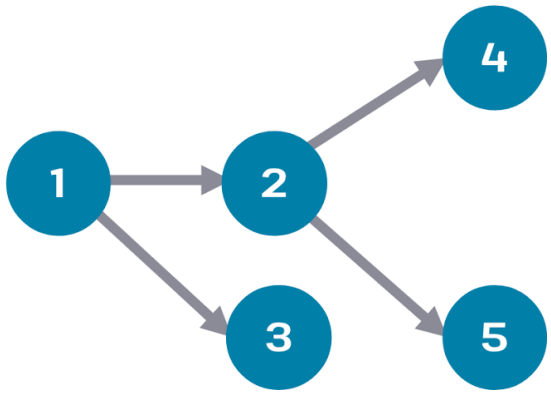
- N positions
- $\leq k$ tiles per position
- $\mathcal{O}(k^2 N)$

Optimal configuration at the MFE

DAG

What is a DAG?

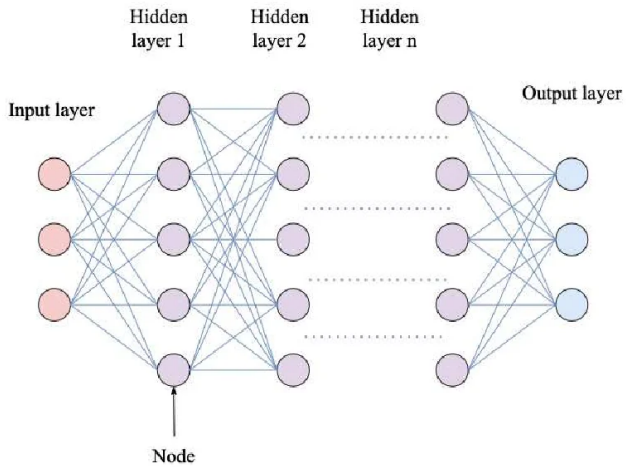
- Nodes
- Directed Edges
- No Cycles



Layered DAG

What is a Layered DAG?

- Directed Acyclic Graph
- Partitioned into Layers
- Topological ordering



Features of a Layered DAG

Why is a Layered DAG relevant to 1D SDC?

- Competing tiles occupy Layers (L_i)
- Layers are ordered ($L_i < L_j$ when $i < j$)
- Edges connect tiles from layer i to layer $i + 1$
- Edge weights - $\Delta G^{\circ}(T, r_i, l_{i+1})$

MFE and Thermodynamic Favourability

Calculating the MFE of a 1D SDC as a Layered DAG traversal problem.

Traversal

What Algorithm can we use to traverse the Layered DAG? And what will a traversal of the Layered DAG give you?

Existing MFE Implementation

How does this approach compare to other approaches, such as the DP approach?

Performance

What is the performance of the shortest path through the Layered DAG?

Conclusion

What are the next steps?